OS & SP

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SKILL – 20

1.Skill20.c

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GNU nano 8.7.1 skill20.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>
#include <termios.h>

#define MAX_ARGS 64

pid_t shell_pgid;
pid_t foreground_pgid = -1;

/* Ctrl+C handler */
void handle_sigint(int sig)
{
    (void)sig;

    if (foreground_pgid > 0) {
        kill(-foreground_pgid, SIGINT);
    }
}

/* Ctrl+Z handler */
void handle_sigtstp(int sig)
{
    (void)sig;

    if (foreground_pgid > 0) {
        kill(-foreground_pgid, SIGTSTP);
    }
}

int main(void)
{
    char command[256];

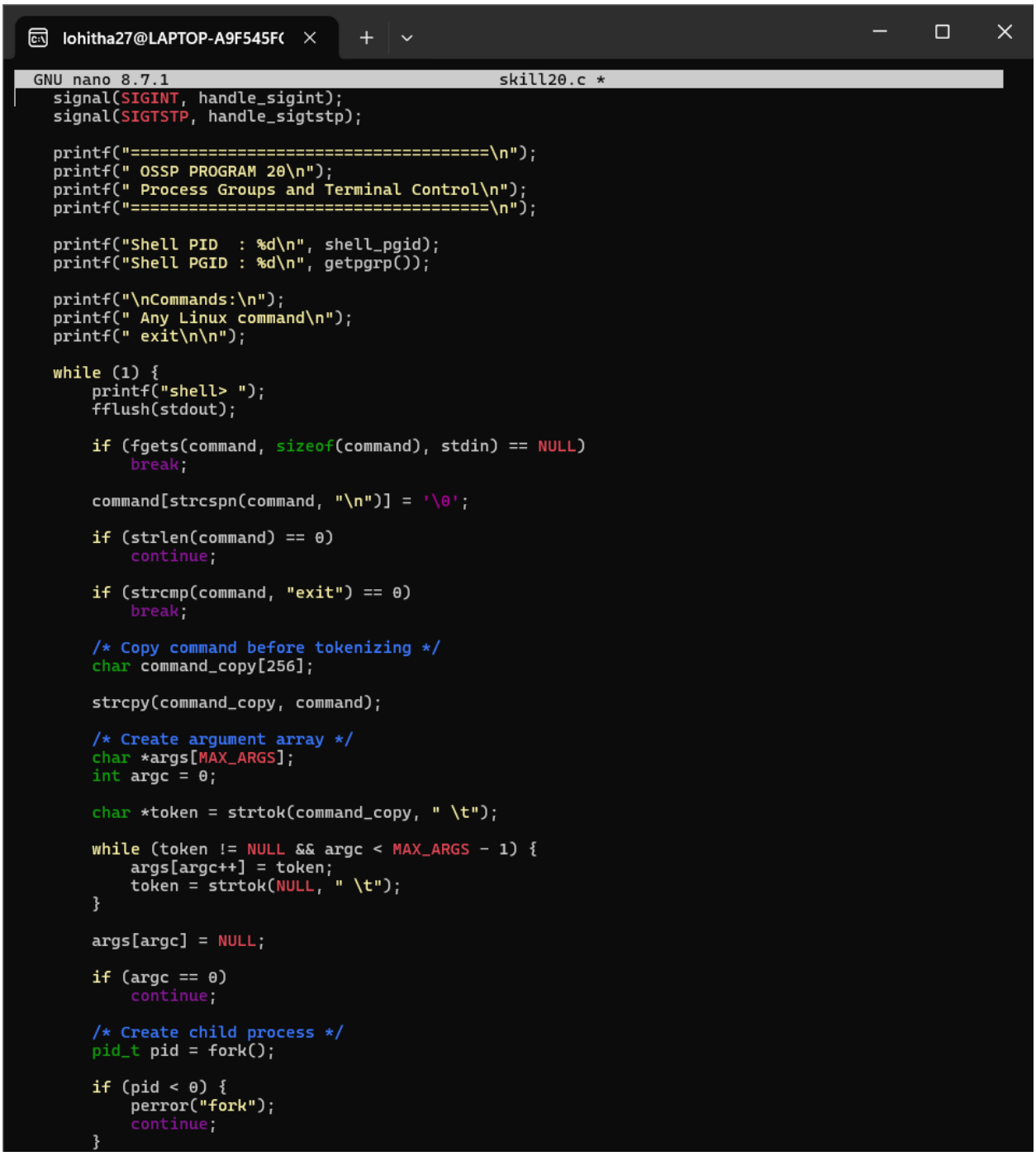
    /* Get shell PID */
    shell_pgid = getpid();

    /* Ignore terminal control signals in shell */
    signal(SIGTTOU, SIG_IGN);
    signal(SIGTTIN, SIG_IGN);
    signal(SIGTSTP, SIG_IGN);

    /* Create shell process group */
    if (setpgid(shell_pgid, shell_pgid) == -1 &&
        getpgrp() != shell_pgid) {
        perror("setpgid shell");
    }

    /* Give terminal to shell */
    if (isatty(STDIN_FILENO)) {
        tcsetpgrp(STDIN_FILENO, shell_pgid);
    }

    /* Register signal handlers */
    signal(SIGINT, handle_sigint);
```


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signal(SIGINT, handle_sigint);
signal(SIGTSTP, handle_sigtstp);
printf("=====\n");
printf(" OSSP PROGRAM 20\n");
printf(" Process Groups and Terminal Control\n");
printf("=====\n");
printf("Shell PID : %d\n", shell_pgid);
printf("Shell PGID : %d\n", getpgrp());
printf("\nCommands:\n");
printf(" Any Linux command\n");
printf(" exit\n\n");
while (1) {
printf("shell> ");
fflush(stdout);
if (fgets(command, sizeof(command), stdin) == NULL)
break;
command[strcspn(command, "\n")] = '\0';
if (strlen(command) == 0)
continue;
if (strcmp(command, "exit") == 0)
break;
/* Copy command before tokenizing */
char command_copy[256];
strcpy(command_copy, command);
/* Create argument array */
char *args[MAX_ARGS];
int argc = 0;
char *token = strtok(command_copy, " \t");
while (token != NULL && argc < MAX_ARGS - 1) {
args[argc++] = token;
token = strtok(NULL, " \t");
}
args[argc] = NULL;
if (argc == 0)
continue;
/* Create child process */
pid_t pid = fork();
if (pid < 0) {
perror("fork");
continue;
}

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```
    }

    /* CHILD */
    if (pid == 0) {
        /* Create a new process group */
        setpgid(0, 0);

        /* Restore default signal behavior */
        signal(SIGINT, SIG_DFL);
        signal(SIGTSTP, SIG_DFL);
        signal(SIGTTIN, SIG_DFL);
        signal(SIGTTOU, SIG_DFL);

        /* Execute command */
        execvp(args[0], args);

        perror("execvp");
        _exit(1);
    }

    /* PARENT */
    else {
        /* Put child in its own process group */
        setpgid(pid, pid);

        foreground_pgid = pid;

        printf("Child PID  : %d\n", pid);
        printf("Child PGID : %d\n", getpgid(pid));
        /* Transfer terminal control to child */
        if (isatty(STDIN_FILENO)) {
            tcsetpgrp(STDIN_FILENO, foreground_pgid);
        }
        int status;
        /* Wait for child to exit or stop */
        if (waitpid(pid, &status, WUNTRACED) == -1) {
            perror("waitpid");
        }
        /* Return terminal control to shell */
        if (isatty(STDIN_FILENO)) {
            tcsetpgrp(STDIN_FILENO, shell_pgid);
        }

        foreground_pgid = -1;

        if (WIFEXITED(status)) {
            printf("Child exited normally.\n");
        }
        else if (WIFSIGNALED(status)) {
            printf("Child terminated by signal %d\n",
                   WTERMSIG(status));
        }
        else if (WIFSTOPPED(status)) {
            printf("Child stopped by signal %d\n",
                   WSTOPSIG(status));
            /* Resume stopped process, as in the source */
            kill(-pid, SIGCONT);
            waitpid(pid, NULL, 0);
```

```
                waitpid(pid, NULL, 0);
            }
        }
    }
    /* Restore terminal to shell */
    if (isatty(STDIN_FILENO)) {
        tcsetpgrp(STDIN_FILENO, shell_pgid);
    }
    printf("\nProgram 20 terminated.\n");
    return 0;
}
```

OUTPUT :

```
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill20.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc -Wall -Wextra -g skill20.c -o skill20
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill20
====================================
 OSSP PROGRAM 20
 Process Groups and Terminal Control
====================================
Shell PID  : 2520
Shell PGID : 2520

Commands:
 Any Linux command
 exit

shell> pwd
Child PID  : 2521
Child PGID : 2521
/home/lohitha27/SKILL
Child exited normally.
shell> ls
Child PID  : 2522
Child PGID : 2522
append.txt    skill10.c  skill13    skill16    skill19    skill3dynamic    skill6    skill9
combined.txt  skill11    skill13.c  skill16.c  skill19.c  skill3dynamic.c  skill6.c  skill9.c
errors.txt    skill11.   skill14    skill17    skill20    skill4           skill7
input.txt     skill11.c  skill14.c  skill17.c  skill20.c  skill4.c         skill7.c
output.txt    skill12    skill15    skill18    skill3     skill5           skill8
skill10       skill12.c  skill15.c  skill18.c  skill3.c   skill5.c         skill8.c
Child exited normally.
shell>
```